Computation for Analytics

DIANE WOODBRIDGE, PHD

Contents

Amazon Web Servies (AWS)

1. Account Setup
2. Add credits and set cost alarm
3. About S3
4. Add data to S3
5. About EC2
6. Launch EC2
7. AWS CLI
8. AWS CLI Install and Configure
9. SSH
10. Run codes on EC2

Contents

Amazon Web Servies (AWS)

1. Account Setup
2. Add credits and set cost alarm
3. About S3
4. Add data to S3
5. About EC2
6. Launch EC2
7. AWS CLI
8. AWS CLI Install and Configure
9. SSH
10. Run codes on EC2

Amazon Web Services (AWS)



It provides scalable and reliable cloud-based computing platforms.

In this program, we will use AWS regularly including in Relational Database, Data Acquisition, Distributed Computing, Distributed Data Systems, Data Science Entrepreneurship, etc.

- We will provide credits for classes only, but it is only $50 or less.
- Make sure to manage your resources to avoid any excessive cost.

https://awsamazoncom/

Contents

Amazon Web Servies (AWS)

1. **Account Setup**
2. Add credits and set cost alarm
3. About S3
4. Add data to S3
5. About EC2
6. Launch EC2
7. AWS CLI
8. AWS CLI Install and Configure
9. SSH
10. Run codes on EC2

AWS Account Set Up

Set up as a non-academic account

- You would need to add your credit card info to move forward.

Contents

Amazon Web Servies (AWS)

1. Account Setup
2. **Add credits and set cost alarm**
3. About S3
4. Add data to S3
5. About EC2
6. Launch EC2
7. AWS CLI
8. AWS CLI Install and Configure
9. SSH
10. Run codes on EC2

Add AWS Credits

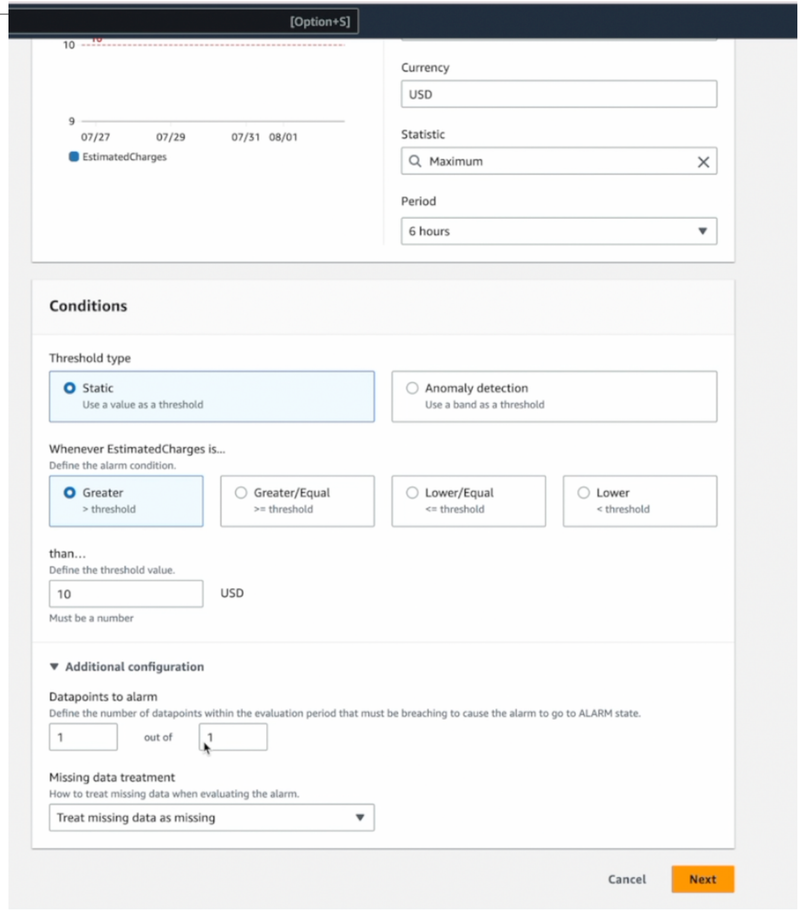
Once account is created, go to Billing Dashboard => Credits and add provided AWS credits.

- This class will provide $50 credits per person and you would need to monitor to avoid any additional cost. (Note: MSDS won’t cover costs more than provided credits)

AWS Billing Alert

In your billing dashboard, go to Billing Preference => Enable CloudWatch billing alerts.

Then go to CloudWatch => Choose the right region (ex. us-east-1), create an alarm with Billing amount as threshold.



https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/monitor_estimated_charges_with_cloudwatch.html

Contents

Amazon Web Servies (AWS)

1. Account Setup
2. Add credits and set cost alarm
3. **About S3**
4. Add data to S3
5. About EC2
6. Launch EC2
7. AWS CLI
8. AWS CLI Install and Configure
9. SSH
10. Run codes on EC2

Amazon Simple Storage Service (S3)



S3 is an object storage that you can store data.

- Object - Data such as files
- Object Keys - A unique identifier for an object
- Bucket - A container for your object
- Bucket Policy - You can provide which users can access your bucket for listing, reading, writing, etc.

https://docs.aws.amazon.com/AmazonS3/latest/

Contents

Amazon Web Servies (AWS)

1. Account Setup
2. Add credits and set cost alarm
3. About S3
4. **Add data to S3**
5. About EC2
6. Launch EC2
7. AWS CLI
8. AWS CLI Install and Configure
9. SSH
10. Run codes on EC2

Add Data to S3

Create S3 bucket and place data

- If needed, you should configure your s3.

Contents

Amazon Web Servies (AWS)

1. Account Setup
2. Add credits and set cost alarm
3. About S3
4. Add data to S3
5. **About EC2**
6. Launch EC2
7. AWS CLI
8. AWS CLI Install and Configure
9. SSH
10. Run codes on EC2

Amazon Elastic Compute Cloud (EC2)



EC2 is a computing platform, where majority of AWS services are running. For instance, AWS EMR (Elastic MapReduce) utilize multiple EC2 instances for creating a cluster and managing resources and computing to enhance scalability.

- Instance - Where your computing is done. You can choose various instance types with different CPU, memory, disk, and networking options.
- You need to configure your instance including user permission and security group to open/close certain port.
- You can ssh (Secure Shell) to the instance.

https://aws.amazon.com/ec2/

Contents

Amazon Web Servies (AWS)

1. Account Setup
2. Add credits and set cost alarm
3. About S3
4. Add data to S3
5. About EC2
6. **Launch EC2**
7. AWS CLI
8. AWS CLI Install and Configure
9. SSH
10. Run codes on EC2

Launch EC2

Choose an instance type including its OS, CPU, memory, disk, etc.

- Make sure to open port 20 to ssh to the instance.
- To SSH, make sure to
  - Create a .pem file and save to your local machine
  - Create IAM user and create its access key (AWS access key and secret access key)

Contents

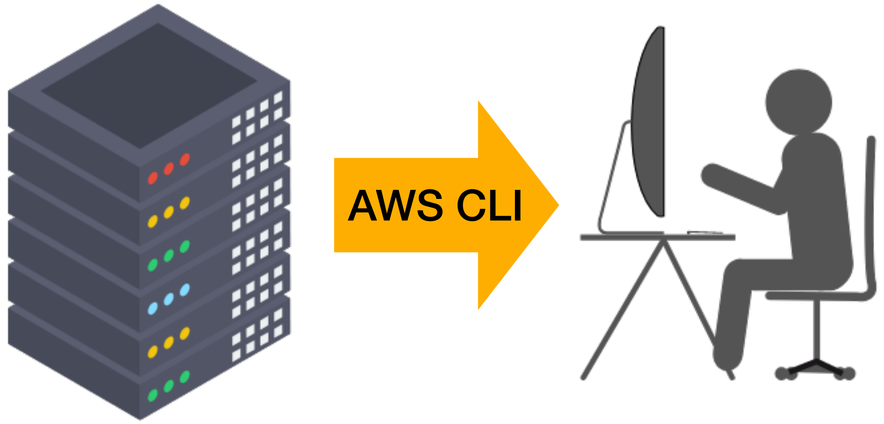
Amazon Web Servies (AWS)

1. Account Setup
2. Add credits and set cost alarm
3. About S3
4. Add data to S3
5. About EC2
6. Launch EC2
7. **AWS CLI**
8. AWS CLI Install and Configure
9. SSH
10. Run codes on EC2

AWS Command Line Interface (CLI)

AWS CLI is a command line tool that you can configure account and access resources provided by AWS.

- Note that AWS CLI includes AWS-specific commands, where you can still use shell commands.



https://aws.amazon.com/cli/
https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html

Contents

Amazon Web Servies (AWS)

1. Account Setup
2. Add credits and set cost alarm
3. About S3
4. Add data to S3
5. About EC2
6. Launch EC2
7. AWS CLI
8. **AWS CLI Install and Configure**
9. SSH
10. Run codes on EC2

AWS CLI Installation and Configuration

Download and install AWS CLI on your local machine.

Configure AWS CLI with the created AWS Access Key and Secret Access Key

- `$aws configure`

Contents

Amazon Web Servies (AWS)

1. Account Setup
2. Add credits and set cost alarm
3. About S3
4. Add data to S3
5. About EC2
6. Launch EC2
7. AWS CLI
8. AWS CLI Install and Configure
9. **SSH**
10. Run codes on EC2

SSH

Secure Shell (SSH)

- Provides a network protocol to communicate between two computers.
- By using SSH, you can login to a server and execute a program on the server.

SSH to EC2

You can SSH to EC2

- Make sure that your .pem file is not accessible by others. (Chmod)
- Commands to SSH

```
aws ec2-instance-connect ssh --instance-id instance_id –private-key-file pem_file_path

or

ssh –i pem_file_path ec2-user@instance_public_ip
```

Contents

Amazon Web Servies (AWS)

1. Account Setup
2. Add credits and set cost alarm
3. About S3
4. Add data to S3
5. About EC2
6. Launch EC2
7. AWS CLI
8. AWS CLI Install and Configure
9. SSH
10. **Run codes on EC2**

Load Data from S3 to EC2

Configure your CLI on EC2 to validate the permission.

Install PIP on EC2

- Then install boto3 via `pip install boto3`

```
import boto3

bucket_name = "usfca-msds501"
key_name = "Data/countries.tsv"

s3 = boto3.client('s3')
object = s3.get_object(Bucket=bucket_name, Key=key_name)
print(object['Body'].read())
```

https://docs.aws.amazon.com/elasticbeanstalk/latest/dg/eb-cli3-install-linux.html
https://aws.amazon.com/sdk-for-python/

Contents

Amazon Web Servies (AWS)

1. Account Setup
2. Add credits and set cost alarm
3. About S3
4. Add data to S3
5. About EC2
6. Launch EC2
7. AWS CLI
8. AWS CLI Install and Configure
9. SSH
10. Run codes on EC2